Predicting Gastrointestinal Disease Risk Using Gut Microbiome Composition: A Machine Learning Approach

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Introduction

This project focused on the role of the human gut microbiome in gastrointestinal (GI) disease risk. This report makes frequent use of microbiology terminology relating to taxonomic classification. For readers not familiar with such terminology, a brief background is provided. Biological taxonomy classifies organisms using seven levels of hierarchical ranking: kingdom, phylum, class, order, family, genus, and species. This project primarily focused on the genus level, as the majority of microbiome studies are able to classify bacteria down to this level (not fully down to the species level). The terms "genera" and "taxa" are used frequently. Genera is the plural form of genus, and taxa is the plural form of taxon (which refers to a single bacterial classification - usually a genus).

The objective of this project was to identify whether the presence of certain bacterial genera in the human gut influences global disease risk. Specifically, the goal was to determine whether veritably beneficial genera (e.g *Lactobacillus*, *Bifidobacterium*) reduce global disease risk, and whether veritably detrimental genera (e.g. *Streptococcus*) increase risk. While these genera are already well-studied, confirming the existing findings through a machine learning approach would be a strong indicator of model robustness. While unsupervised learning may reveal interesting insights involving microbial correlation as well as synergistic behavior between bacterial groups (genera or otherwise), the questions driving this project were predictive in nature. As such, this project employed supervised learning, structuring the challenge of disease state prediction using binary classification.

This subject matter is of particular interest to me, both academically and personally. Microbiome therapeutics is a very young, yet promising, field of research, and I believe that the coming decades will bring entirely new therapeutic modalities targeting the microbiome for treating both physical and mental illnesses. I have personally explored such approaches for my own well-being and have observed unexpectedly positive outcomes. This project represented an opportunity to further my understanding of the field overall, as well as explore how gut flora can influence systemic health. The findings were translated into practical approaches that could potentially guide individualized therapeutic strategies.

Approach

<u>Problem Framing</u>

The central question driving this research was as follows: Does the unique composition of an individual's gut microbiome, in combination with patient characteristics like BMI, age and geographical location, play a role in driving disease risk?

The generating process comprises the development of disease (any GI disease, in this project) as a function of an individual's microbiome composition and additional patient characteristics. The data used for this project effectively modeled this generating process by representing microbiome composition using microbial abundance (16S rRNA amplicon sequencing) as well as supplemental tags and metadata describing patient disease states. The data provided the necessary components for modeling the probability of disease presence given these patient-level inputs.

Binary cross-entropy served as the objective function throughout this project, given that global disease state prediction (diseased or non-diseased) is a binary classification task. The chosen models were optimized to predict disease state using the available input features (initially planned to include microbial abundances, BMI, age, and geographical region, though patient metadata was ultimately excluded as described below).

Data Framing

The dataset used for this project was sourced from the Human Microbiome Project (HMP), a comprehensive repository of 168,464 samples aggregated from 482 microbiome studies. The HMP granted this project a much larger scope than would be possible otherwise, however it also introduced several challenges stemming from the diversity of studies included in the dataset.

Available features included:

- Raw read counts of bacterial taxa: The same 4680 bacterial taxa were sequenced for each patient in the dataset using the same pipeline and reference database (critical for comparability between samples).

- Patient tags: Certain metadata such as age, body mass index (BMI), gender, geographic region, and sample date were available for all samples. However, due to the diversity of included studies, many tags were study-specific (e.g. a depression study includes several unique tags describing medication usage, anxiety level, mood, etc. that are not present in samples from other studies).

Initially, feature selection was planned to prioritize patient characteristics that were available across as many studies as possible, such as BMI, age, and region. However, during data exploration, it was discovered that BMI and age were missing from approximately 50% of the samples in the filtered GI disease dataset. Including these features would have required either substantial imputation (which could introduce bias) or discarding half the dataset. Given the importance of sample size for model robustness, the decision was made to exclude BMI and age from the final training set.
Geographic region was also evaluated as a potential feature. While region data was more complete, initial experiments revealed that it was highly imbalanced across the dataset and did not improve model performance. Consequently, region was also excluded from the final dataset.

Available labels included disease status, however the specific format and terminology varied drastically from study to study. Certain studies used a severity scale which varied per study (e.g. 0-10 in some studies, 1-5 in others), while others used a binary measurement that also varied per study ("I have/do not have this disease" in some studies, 1/0 in others).

Two strategies were considered for structuring the binary classification task. The first option involved filtering the dataset to one specific disease, allowing for a more granular analysis while sacrificing most of the data. The second option involved pooling each distinct disease into one unified binary label, drastically increasing sample size while sacrificing specificity. This project applied the second approach, as it aligned with the project's primary objective: understanding to what extent microbiome composition drives global disease risk. Furthermore, many GI diseases like irritable bowel syndrome (IBS), inflammatory bowel disease (IBD), and dyspepsia share overlapping symptoms and, while research is still inconclusive, are believed to originate from a similar microbial imbalance (dysbiosis). Therefore, modelling global GI disease risk rather than focusing on individual diseases could still provide practical and generalizable insights.

<u>Objective Framing</u>
While simple accuracy is an intuitive choice for a performance metric, it can be misleading for datasets with significant class imbalance. A more robust performance metric is the area under the receiver operating characteristic curve (ROC AUC), which was employed for all models evaluated in this project. Precision, recall, specificity and confusion matrix analysis were also employed for model evaluation. In summary, binary cross-entropy was minimized during training, and both ROC AUC and confusion matrices were used for model comparison.

<u>Data Acquisition and Cleaning</u>
The HMP dataset (microbiomap.org) provided an aggregate dataset from 482 microbiome studies. The full dataset comprised 168,464 samples, with each sample accompanied by the read counts (abundances) of 4,687 distinct taxonomic features. The dataset was divided into three tables:

1. The taxonomic table (a matrix of samples and taxon features)
2. Metadata (sample-level information)
3. Tags (sample tags specifying disease states, patient characteristics like BMI, among others that vary between studies)

The dataset was cleaned to unify any discrepancies between study reporting styles. For example, certain studies reported disease severity using a scale, while others reported binary disease presence. Furthermore, certain studies used the "Tag" and "Value" columns present in the "Tags" table differently. Some studies utilized the "Tag" column for specific disease names (e.g. "IBS"), and the "Value" column for disease state (e.g. 1 or 0, "I have/do not have this", or some form of severity scale that varied between studies). Other studies, however, specified the literal text "Disease Name" in the "Tag" column, and specified the actual disease name in the "Value" column if it was present. Each of these discrepancies was unified into one cohesive label before the data was used for model training.

Final Dataset Statistics

Total samples	11,586
Healthy samples	7,442
Diseased samples	4,144
Class balance	0.358 (diseased/total)
Train/Test Split	75/25 (8689/2897)

Feature Preparation and Normalization
The HMP dataset provided only raw read counts of each taxon. Direct comparison of raw counts is meaningless, as sequencing depth (the richness/concentration of the sample being sequenced) varies across samples. To correct for these differences, three steps were applied:

1. Total sum scaling (TSS) to convert raw counts to relative abundances per sample.
2. Centered log-ratio (CLR) to mitigate the inherent dependence present when representing taxon features using relative abundance. CLR transformation is essential for compositional data, as it accounts for the fact that an increase in one taxon's relative abundance necessarily causes a decrease in others.
3. Standard scaling to normalize feature variance for machine learning compatibility.

Taxon features with exclusively zero values ("dead features") or insufficient specificity (e.g. taxa that were only identified to the class or order level) were removed. Features identified to at least the family or genus level were retained.

Final Feature Set
Original features: 4,680 bacterial taxa
After filtering shallow taxa (< family level): ~3,200 taxa
After removing dead features: 2,597 bacterial taxa

Methodology and Models

Baseline Model 1: Logistic Regression
Given the binary nature of this project's research question, logistic regression served as a baseline model. Logistic regression provides highly interpretable feature coefficients, with the magnitude and direction of each coefficient indicating feature importance in predicting disease likelihood.

Following consultation with a machine learning colleague experienced in metabolomics and microbiome research, ElasticNet regularization was employed for the logistic regression model.

ElasticNet is standard practice in microbiome research for its ability to handle correlated features (common in compositional microbiome data) while maintaining interpretability. The model was optimized using Bayesian hyperparameter search with 5-fold stratified cross-validation.

Performance: The logistic regression model achieved a cross-validation ROC AUC of 0.7796 and demonstrated strong interpretability through its feature coefficients.

Baseline Model 2: XGBoost

Ensemble models (such as gradient boosted trees) are particularly well-suited to sparse data like the dataset used in this project. XGBoost, while more complex than simple logistic regression, remained interpretable using SHAP (Shapley Additive Explanations) values.

The XGBoost model was optimized using Bayesian hyperparameter search with 5-fold cross-validation across the following parameters: learning rate, max depth, number of estimators, subsample ratio, column sample ratio, and L1/L2 regularization.

Performance: The XGBoost model achieved a cross-validation ROC AUC of 0.8394, representing strong predictive performance.

Additional Model: LightGBM

Following the same consultation with a machine learning colleague, LightGBM was added as an additional model. LightGBM is commonly used in metabolomics and microbiome studies due to its efficiency with high-dimensional, sparse data and its ability to handle the compositional nature of microbiome features effectively.

Similar to XGBoost, LightGBM was optimized using Bayesian hyperparameter search with 5-fold stratified cross-validation. The model architecture included leaf-wise tree growth (as opposed to XGBoost's level-wise growth), which is particularly effective for datasets with many features.

Performance: The LightGBM model achieved a cross-validation ROC AUC of 0.8340, nearly matching the performance of XGBoost with a significantly faster training time.

Advanced Model: Denoising Autoencoder + Classifier

A neural network served as a more complex model, allowing for nonlinear relationships between taxon features to be captured. Microbes do not exist in isolation within the gut, and their synergistic relationships (rather than purely their individual identities) may be just as important to consider in the context of disease prediction. For example, one genus of gut bacteria may metabolize dietary fiber into some metabolite that is biologically irrelevant to the human host, while another may further process this metabolite into a biologically active product (e.g. short-chain fatty acids), that is highly protective to the host (thus reducing disease risk). Capturing these intricate relationships between taxa required a more complex model architecture.

While neural networks offer a deeper understanding of how gut microbes interact synergistically, they are less robust to sparse data. To address this challenge, a denoising autoencoder (DAE) was implemented to learn a compressed, latent representation of microbial taxon features and improve robustness.

The DAE architecture comprised:

- Encoder: 2,597 input features -> 256 neurons (ReLU) -> 128 neurons (ReLU) -> 512 neurons (ReLU, encoding layer)
- Decoder: 512 neurons -> 128 neurons (ReLU) -> 256 neurons (ReLU) -> 2,597 neurons (linear output)
- Noise factor: 0.2 (Gaussian noise added to input during training)
- Compression ratio: 5.1x (2,597 features compressed to 512)

The autoencoder was trained for 50 epochs to reconstruct clean input from noisy input. Following encoding, a supervised classifier was trained on the compressed 512-dimensional representation:

- Input: 512 encoded features
- Hidden layer: 32 neurons (ReLU) with 30% dropout
- Output: 1 neuron (sigmoid activation)

The classifier was trained with early stopping (patience: 10 epochs, monitoring validation AUC) to prevent overfitting.

Performance: The DAE + classifier model achieved a test set ROC AUC of 0.6945.

Summary of Results

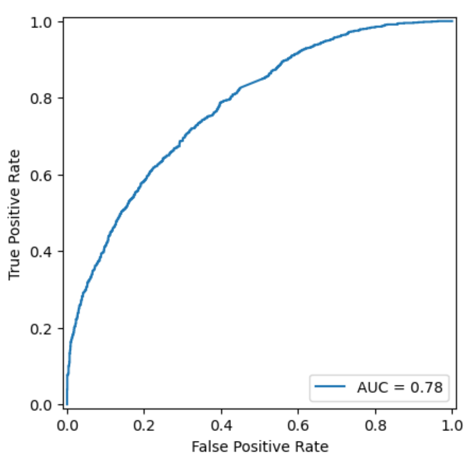
ElasticNet Logistic Regression

Protective Genera via Direct Coefficient Interpretation				
Rank	**Genus**	**Family**	**Coef.**	**Interpretation**
1	*Oxidoreducens group*	-	-0.2583	Strongest protective effect
2	*Lachnotalea*	Lachnospiraceae	-0.2315	Known butyrate producer (a healthy metabolite)
3	*Tumebacillus*	Alicyclobacillaceae	-0.2279	Protective
4	*Oscillatoria*	Desertifilaceae	-0.2163	Protective
5	*Incertae sedis*	-	-0.151	Protective
6	*Pantoea*	Erwiniaceae	-0.1353	Protective
7	*Pseudoramibacter*	Eubacteriaceae	-0.1346	Protective
8	*Genus 10*	-	-0.1245	Protective

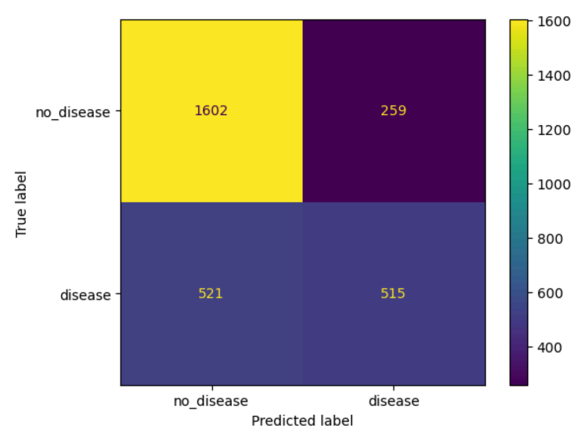
9	*Salmonella*	Enterobacteriaceae	-0.1154	Protective (highly unexpected)
10	*Bradyrhizobium*	Xanthobacteraceae	-0.1101	Protective

Risk-Enhancing Genera via Direct Coefficient Interpretation				
Rank	**Genus**	**Family**	**Coef.**	**Interpretation**
1	*Anoxybacillus*	Bacillaceae	0.2071	Strongest risk-enhancing
2	*Weizmannia*	Bacillaceae	0.155	Risk-enhancing
3	*Leptotrichia*	Leptotrichiaceae	0.1547	Risk-enhancing
4	*Paucilactobacillus*	Lactobacillaceae	0.1508	Risk-enhancing (unexpected for Lactobacillus family)
5	*Cutibacterium*	Propionibacteriaceae	0.1278	Associated with inflammation (causes acne)
6	*Tissierella*	-	0.1216	Risk-enhancing
7	*Massilia*	Oxalobacteraceae	0.1145	Risk-enhancing
8	*Mucilaginibacter*	Sphingobacteriaceae	0.1096	Risk-enhancing
9	*Mammaliicoccus*	Staphylococcaceae	0.1019	Risk-enhancing
10	*Herbaspirillum*	Oxalobacteraceae	0.1012	Risk-enhancing

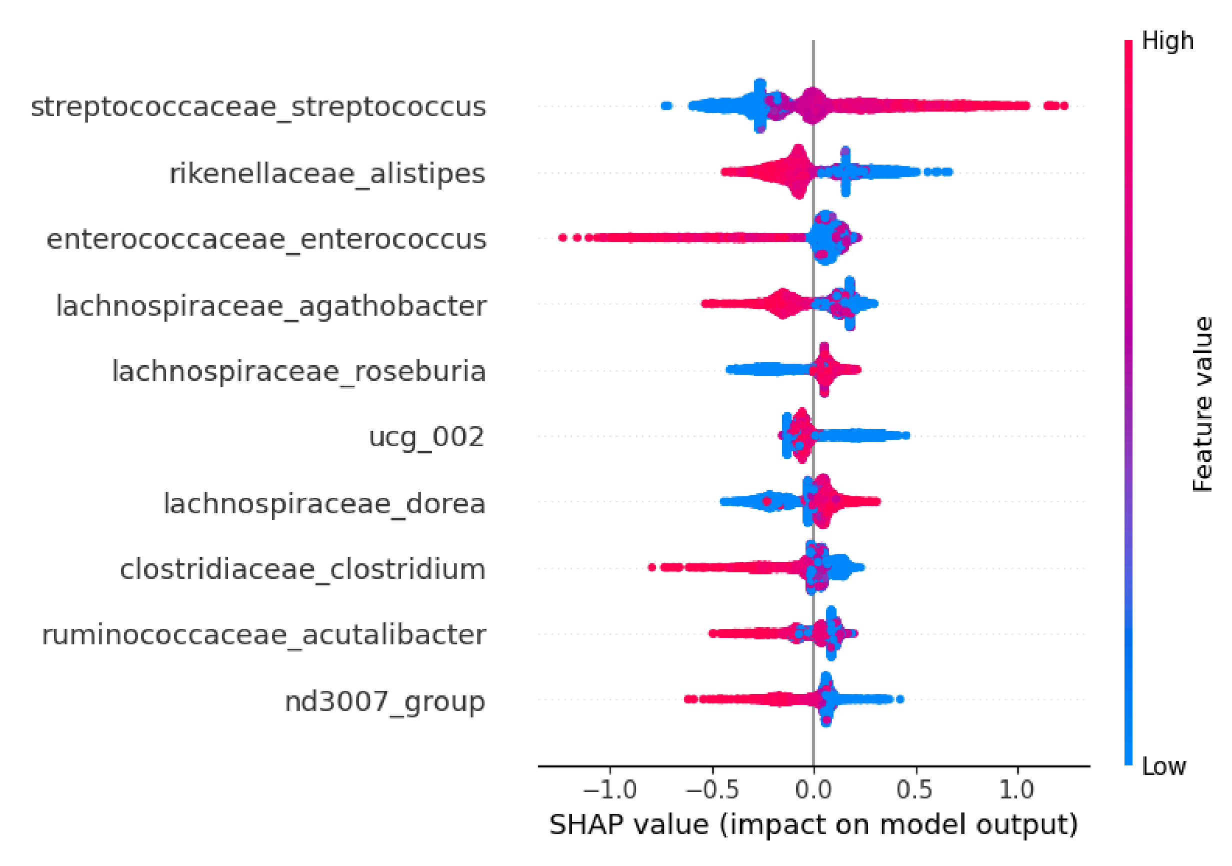
ROC AUC



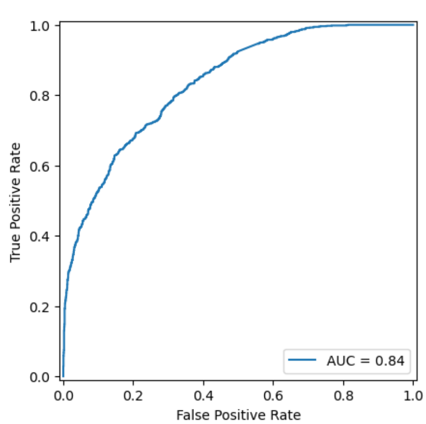
Confusion Matrix



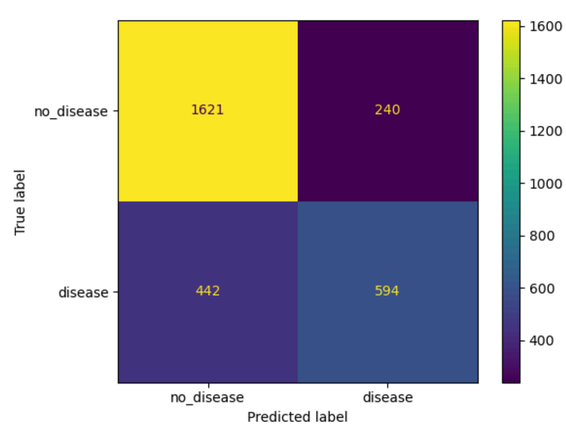
XGBoost



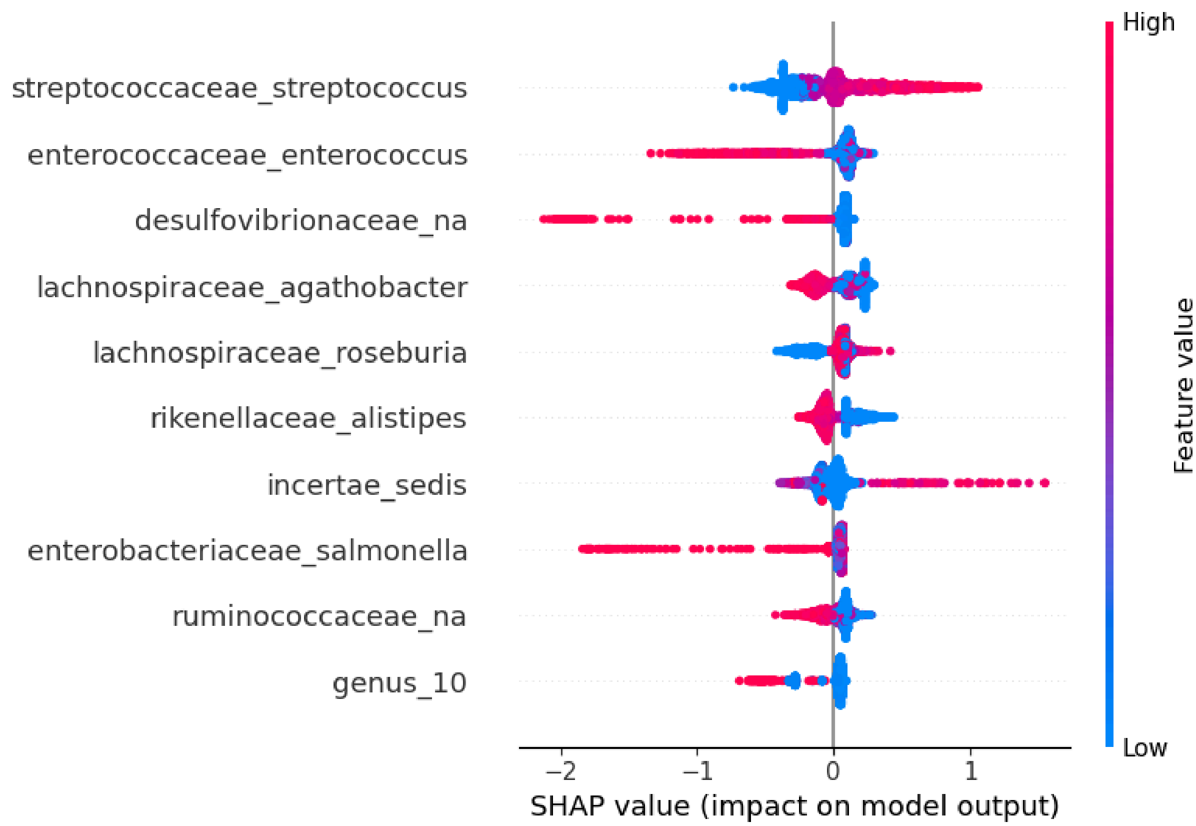
ROC AUC

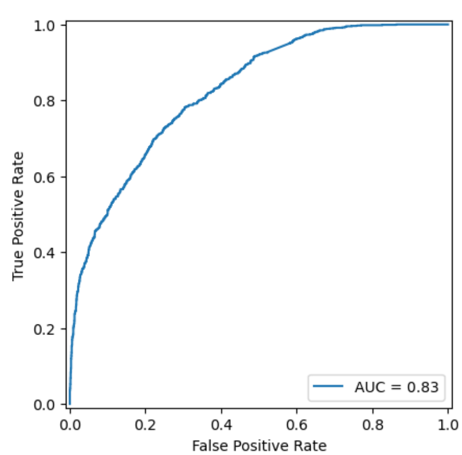
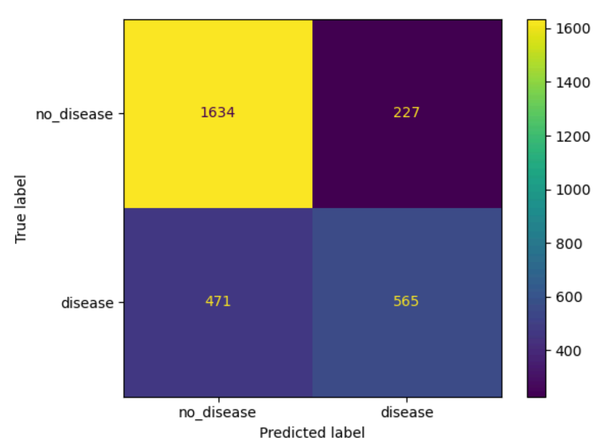


Confusion Matrix

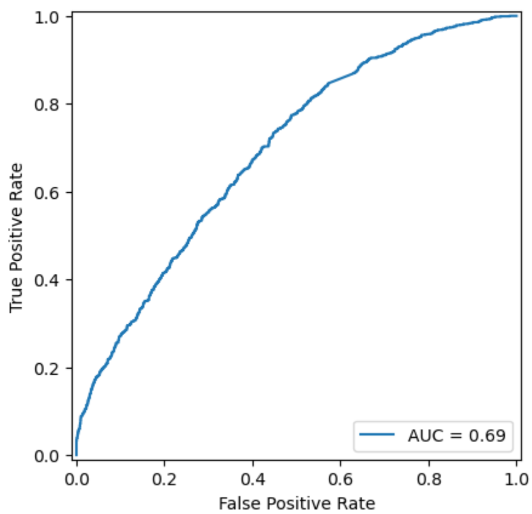
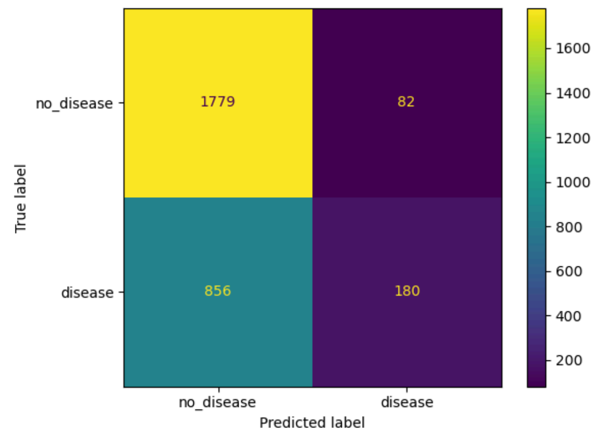


LightGBM



ROC AUC	Confusion Matrix
	

DAE + Neural Network

ROC AUC	Confusion Matrix
	

Note on Feature Importance Consistency

SHAP analysis was conducted for XGBoost and LightGBM models to identify the most influential bacterial genera for disease prediction. While the exact rankings varied slightly between models, there was notable consistency in which genera appeared among the top influential features across multiple models (*Streptococcus*, *Enterococcus*, *Agathobacter*, *Roseburia*, and *Alistepes*). This cross-model agreement provides confidence that the identified bacterial taxa represent genuine biological signals rather than model-specific artifacts. *Streptococcus*, in particular, was highly anticipated as an influential genus as mentioned in the project's hypothesis. Its presence as the most risk-enhancing genus in both tree models was an exciting observation.

The direct coefficient interpretations of the ElasticNet logistic regression model did not align with the SHAP analysis of the tree ensembles. However, coefficients and SHAP values cannot be compared directly, and this discrepancy is not unexpected. Coefficients represent the change in log-odds per unit increase while other coefficients are held constant, while SHAP values represent the average marginal contributions of features to model predictions compared to a baseline. The DAE + neural network approach was not included in feature importance comparison, as its poor performance did not warrant further analysis.

Model Comparisons

Model	CV ROC AUC	Test ROC AUC	Precision	Recall	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)
XGBoost	0.8394	0.8353	0.7122	0.5734	1621	240	442	594
LightGBM	0.834	0.8277	0.7134	0.5454	1634	227	471	565

ElasticNet Log. Reg.	0.7796	0.7775	0.6654	0.4971	1602	259	521	515
DAE + Classifier	-	0.6945	0.687	0.1737	1779	82	856	180

Metric	Best Model	Value	Interpretation
Highest Test ROC AUC	XGBoost	0.8353	Best overall discriminative performance
Highest CV ROC AUC	XGBoost	0.8394	Most robust to cross-validation variation
Highest Precision	LightGBM	0.7134	Fewest false positives among positive predictions
Highest Recall	XGBoost	0.5734	Best at identifying true disease cases
Highest Specificity	DAE + Classifier	0.9559	Best at correctly identifying healthy cases
Most Efficient	LightGBM	0.8277 ROC AUC	Great performance with fastest training time

XGBoost demonstrates the strongest overall discriminative performance, achieving the highest CV ROC AUC (0.8394) and test ROC AUC (0.8353), reflecting both cross-validation stability and reliable generalization. Its precision (0.7122) indicates the model was able to identify a significant proportion of true positives among predicted positives. Its recall (0.5734) shows it is the most effective model at identifying true disease cases, however this value indicates that a large proportion of true disease cases remain undetected. LightGBM closely follows in test ROC AUC (0.828) and achieves the highest precision (0.7134), while also offering a more efficient training time. ElasticNet logistic regression exhibits reduced discriminative ability (test ROC AUC 0.7775) and lower recall, leading to more false negatives, but remains the most interpretable model. In contrast, the DAE + classifier achieves very high specificity (0.9559), but this comes at the cost of substantially weaker overall discrimination (ROC AUC 0.695), limiting its practical utility. This architecture likely performed poorly due to insufficient tuning. More rigorous hyperparameter tuning (using Bayes search CV or another approach) may be required to enable greater discriminative performance. Overall, XGBoost provides the best balance across performance metrics, and LightGBM is preferable when precision and training efficiency are prioritized.

Biological Interpretation

The logistic regression coefficients revealed interesting patterns that both align with and challenge existing microbiome research. While the original hypothesis predicted that well-known beneficial genera like *Lactobacillus* and *Bifidobacterium* would show protective effects, these genera were not among the top protective features identified. Instead, Lachnospiraceae members (such as *Lachnotalea*), which are known producers of short-chain fatty acids (SCFA) like butyrate, emerged as strongly protective. This finding is consistent with the established role of butyrate in maintaining gut barrier integrity and reducing inflammation.

Interestingly, some risk-enhancing genera were unexpected. For instance, Lactobacillaceae *Paucilactobacillus* (a *Lactobacillus*-related genus) showed a positive coefficient (0.151), suggesting increased disease risk rather than the hypothesized protective effect. This may highlight the importance of species- and strain-level specificity in microbiome research (not all members of traditionally "beneficial" families necessarily provide health benefits.)

The protective effect of Enterobacteriaceae *Salmonella* (coefficient: -0.115) was also surprising and warrants further investigation, as this finding may reflect confounding factors in the aggregated disease dataset. It's possible that in certain disease contexts, the presence of specific taxa reflects compensatory responses rather than causal factors. For example, high prevalence of certain pathogenic taxa in severe dysbiotic states may result in a reduction in *Salmonella* through competitive exclusion. As a result, the model may have learned that *Salmonella* presence implies a less severe dysbiotic state.

Perhaps the most exciting validation of tree ensemble robustness is the consistent identification of *Streptococcus* as the most detrimental genera during SHAP analysis. As mentioned in the project hypothesis, *Streptococcus* is a well-established driver of disease extending beyond the GI tract, and its prominence here is a fantastic confirmation of model performance. Furthermore, the SHAP ranking of *Agathobacter* and *Roseburia* as highly protective offers further support. These are both well-studied, beneficial, SCFA-producing taxa, and their reproducible importance across models increases the biological plausibility of these results.

Building on these findings, future work should explore the pathological significance of other consistently high-ranking genera identified by the models, with particular emphasis on determining whether their effects are amenable to targeted intervention. Reframing the problem to more fully leverage the entirety of the original Human Microbiome Project dataset rather than collapsing outcomes into a single disease label could also enable multilabel classification approaches to predict specific disease risk. Additionally, extending this modeling framework to the rapidly developing field of gut–brain axis research offers a promising means of understanding how microbial composition relates to mental health, with the longer-term goal of improving prediction and treatment of mental illness using microbial abundance profiles and rational probiotic supplementation. Finally, applying similar models to predict host BMI might provide insight into how body fat and metabolism are related to gut microbial composition.

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